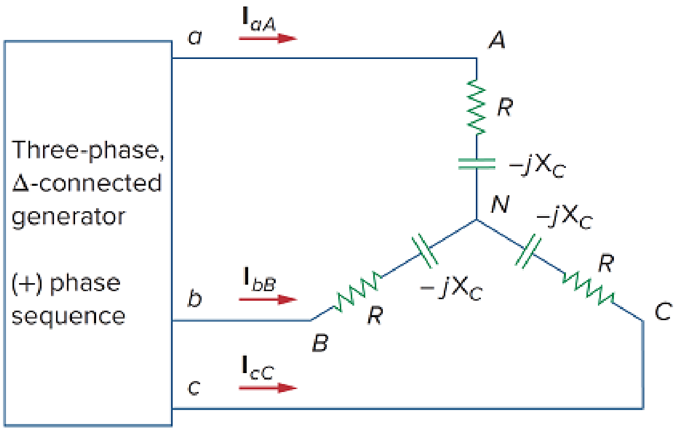
Chapter 12, Problem 26P

Problem

Using Fig. 12.55, design a problem to help other students better understand balanced delta connected sources delivering power to balanced wye connected loads.

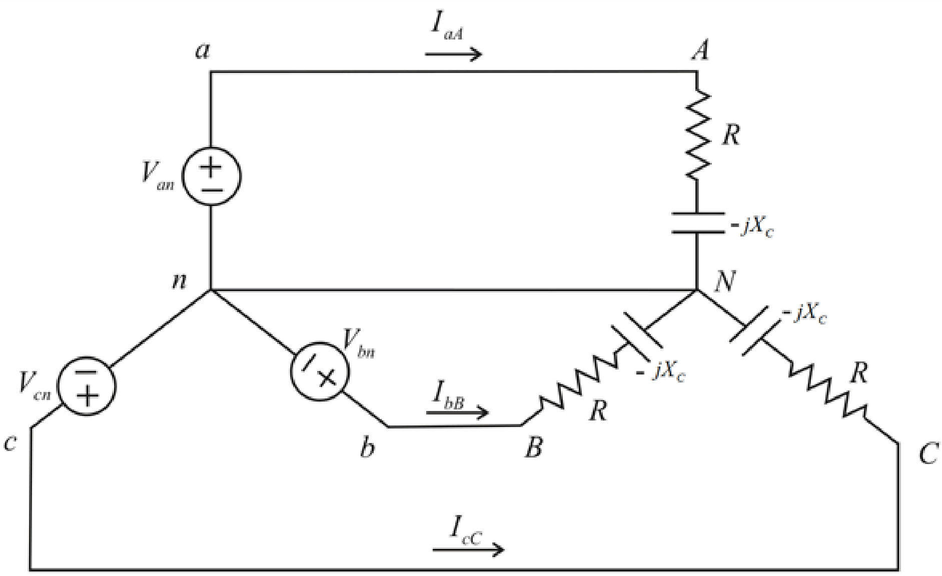
Figure 12.55



Step-by-step solution

Step 1 of 2

We first convert the delta to wye as shown in below circuit,



Step 2 of 2

Where,

$$V_{an} = \frac{V_{ab}}{\sqrt{3}\angle 30^\circ}$$

$$V_{an} = \frac{V_{ab}\angle -30^\circ}{\sqrt{3}}$$

Therefore the line currents are,

$$I_{aA} = \frac{V_{an}}{R - jX_C}$$

$$\therefore \boxed{I_{aA} = \frac{V_{ab}\angle -30^\circ}{\sqrt{3}\left(R - jX_C\right)}}$$

$$I_{bB} = I_{aA}\angle -120^\circ$$

$$I_{bB} = \frac{V_{ab}\angle -30^\circ}{\sqrt{3}\left(R - jX_C\right)} \times \left(1\angle -120^\circ\right)$$

$$\therefore \boxed{I_{bB} = \frac{V_{ab}\angle -150^\circ}{\sqrt{3}\left(R - jX_C\right)}}$$

$$I_{cC} = I_{aA}\angle 120^\circ$$

$$I_{cC} = \frac{V_{ab}\angle -30^\circ}{\sqrt{3}\left(R - jX_C\right)} \times \left(1\angle 120^\circ\right)$$

$$\therefore \boxed{I_{cC} = \frac{V_{ab}\angle 90^\circ}{\sqrt{3}\left(R - jX_C\right)}}$$